

Autumn Crossword Solutions

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1 3	2 1	3 6
4 9	3 3	8 8
5 9	6 6	0 0

ONE ACROSS

Let $A = 3a + 1$ and $B = 3b + 1$, since $A, B \equiv 1 \pmod{3}$, and let $C = 3c$ and $D = 3d$, since $C, D \equiv 0 \pmod{3}$. The bounds give $0 \leq a, b \leq 7$ and $0 \leq c, d \leq 9$. From $A + B + C + D = 35$, we get $(3a + 1) + (3b + 1) + 3c + 3d = 35$, so $a + b + c + d = 11$. Ignoring the upper bounds, the number of nonnegative integer solutions is $\binom{14}{3} = 364$.

Now subtract solutions exceeding the bounds. If $a \geq 8$, setting $a' = a - 8$ gives $a' + b + c + d = 3$, so there are $\binom{6}{3} = 20$ such solutions; the same holds for b , giving 40 total. If $c \geq 10$, setting $c' = c - 10$ gives $a + b + c' + d = 1$, so there are $\binom{4}{3} = 4$ such solutions; similarly for d , giving 8 total. No two violations can occur simultaneously since even $8 + 8 > 11$.

Therefore, the answer is $364 - 40 - 8 = \boxed{316}$.

ONE DOWN

We use the Principle of Inclusion-Exclusion to find the total assignments of 8 people to 3 distinct non-empty groups:

$$3^8 - 3(2^8) + 3(1^8) = 6561 - 768 + 3 = 5796.$$

Then, we calculate the restricted cases where Alice and Bob are grouped together, treating them as 1 bundled entity (leaving 7 total entities):

$$3^7 - 3(2^7) + 3(1^7) = 2187 - 384 + 3 = 1806.$$

Subtracting the restricted cases from the total, we then arrive at $N = 5796 - 1806 = 3990$, so $N/10 = \boxed{399}$.

TWO DOWN

Let $N = R + G$. Since

$$\frac{\binom{R}{2} + \binom{G}{2}}{\binom{N}{2}} = \frac{1}{2},$$

we get $4RG = N(N - 1)$. Thus,

$$(R - G)^2 = (R + G)^2 - 4RG = N,$$

so N must be a perfect square. Between 240 and 260, the only perfect square is 256. Therefore, $|R - G| = 16$, and to maximize R , we take $R - G = 16$ to get

$$R = \frac{256 + 16}{2} = \boxed{136}.$$

THREE DOWN

Calculating the first few terms ($A_2 = 3, A_3 = 6, A_4 = 10$) reveals that A_n is the n -th triangular number, $A_n = \frac{n(n+1)}{2}$, which can be verified through induction. Summing the first 15 terms, we then have

$$\begin{aligned}\sum_{n=1}^{15} A_n &= \sum_{n=1}^{15} \frac{n^2 + n}{2} = \frac{1}{2} \left(\sum_{n=1}^{15} n^2 + \sum_{n=1}^{15} n \right) \\ &= \frac{1}{2} \left(\frac{15(16)(31)}{6} + \frac{15(16)}{2} \right) \\ &= \frac{1}{2} (1240 + 120) = \boxed{680}.\end{aligned}$$

FOUR ACROSS

Let $P(x) = Ax^3 + Bx^2 + Cx + D$. Then, $P(x+1) - P(x) = A(3x^2 + 3x + 1) + B(2x + 1) + C$. Comparing coefficients with $3x^2 - 143x + 396$ gives $3A = 3$, $3A + 2B = -143$, and $A + B + C = 396$, so $A = 1$, $B = -73$, and $C = 468$. Thus, $P(x) = x^3 - 73x^2 + 468x + D$. Since $P(2) = 256$, we get $8 - 292 + 936 + D = 256$, so $D = -396$.

Therefore, $P(x) = x^3 - 73x^2 + 468x - 396$. Since its roots are a, b, c , we have $P(x) = (x - a)(x - b)(x - c)$. Substituting $x = -1$ gives

$$P(-1) = (-1 - a)(-1 - b)(-1 - c) = -(a + 1)(b + 1)(c + 1).$$

Also,

$$P(-1) = -1 - 73 - 468 - 396 = -938,$$

$$\text{so } (a + 1)(b + 1)(c + 1) = \boxed{938}.$$

FIVE ACROSS

Place H at 0 on BC . Since $AH = 24$ and $AM = 6\sqrt{41}$, right triangle AHM gives

$$HM = \sqrt{AM^2 - AH^2} = \sqrt{36(41) - 576} = 30.$$

Let $BH = x$, so because M is the midpoint of BC , we have $HC = x + 60$. Also $HD = 6$, so $BD = x + 6$. Let $AB = \sqrt{x^2 + 576}$ and $AC = \sqrt{(x + 60)^2 + 576}$. Moreover, since $BD = \frac{AB + BC - AC}{2}$, we have

$$x + 6 = \frac{AB + (2x + 60) - AC}{2} \Rightarrow AC - AB = 48.$$

Thus,

$$\sqrt{(x + 60)^2 + 576} - \sqrt{x^2 + 576} = 48.$$

Writing $\sqrt{x^2 + 576} = y$, squaring once gives $5x + 54 = 4y$, so $16(x^2 + 576) = (5x + 54)^2$. This simplifies to $x^2 + 60x - 700 = 0$, giving $x = 10$ since $x > 0$. Hence, $BC = 2x + 60 = 80$, and so the area is $\frac{1}{2}(80)(24) = \boxed{960}$.